$$= \frac{1}{2}\ [\frac{8}{3} - 0\ ]$$

$$= \frac{4}{3}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

$$= \int_{-\infty}^{\infty} x^2 \left(\frac{x}{2}\right) dx$$

$$= \frac{1}{2} \left[x^4/4\right]_0^2$$

$$= \frac{1}{8}\ [16\text{-}0]$$

$$= 2$$

$$\text{Variance } (X) = E(X^2) - [E(X)]^2$$

$$= 2 - (\frac{4}{3})^2$$

$$= 2 - \frac{16}{9} = \frac{2}{9}$$

NOTE

The following results are true in both discrete and continuous cases.

(i) $E\ (\ 1/X) \neq 1/\ E(X)$

(ii) $E\ [\log\ (X)] \neq \log E\ (X)$

(iii) $(E\ (X^2) \neq [E(X)]^2$

So far we have studied how to find mean and variance in the case of single random variables taken at a time but in some cases we need to calculate the expectation for a linear combination of random variables like $aX + bY$ or the product of the random variables $cX \times dY$ or involving more number of random variables. So here we see theorems useful in such situations.

9.6.3 Independent random variables

Random variables X and Y are said to be independent if the joint probability density function of X and Y can be written as the product of marginal densities of X and Y.

That is $f\ (x,y) = g(x)\ h(y)$

Here $g(x)$ marginal p.d.f. of X

$h(y)$ marginal p.d.f. of Y

9.7 Addition and Multiplication Theorem on Expectations

9.7.1 Addition Theorem on Expectations

1. Statement for Discrete random variable

If X and Y are two discrete random variables then

$$E\ (X+Y) = E(X) + E(Y)$$